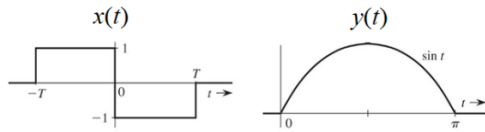


ENGR 3323: Signals and Systems

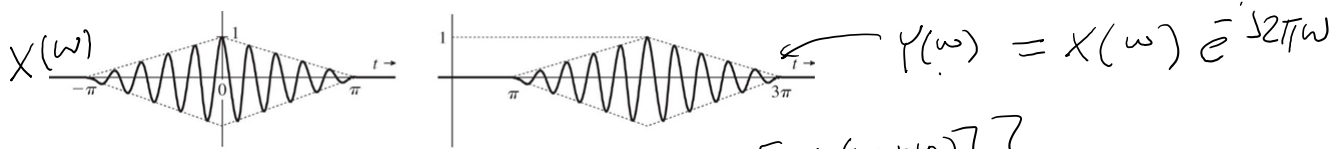
HW 11_Ch7

Q1) Find the Fourier transforms (FT) of the signals shown below using only the time-shifting property and the FT Table.



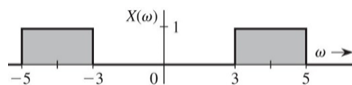
$$X(\omega) = \frac{j4}{\omega} \sin^2\left(\frac{\omega T}{2}\right) \quad Y(\omega) = \frac{1}{1-\omega^2} (1 + e^{-j\pi\omega})$$

Q2) The signals below are modulated signals with carrier $\cos(10t)$. Find the Fourier transforms of these signals by using the Fourier transform properties and the FT table. Sketch the amplitude and phase spectra for the signals.



$$X(\omega) = \frac{\pi}{2} \left\{ \sin^2 \left[\frac{\pi(\omega - 10)}{2} \right] + \sin^2 \left[\frac{\pi(\omega + 10)}{2} \right] \right\}$$

Q3) Use the frequency-shifting property and the FT Table to find the inverse Fourier transform of the spectra shown below



$$x(t) = \frac{2}{\pi} \text{sinc}(t) \cos 4t$$

Q4) For a stable LTIC system with transfer function $H(s) = 1/(s + 1)$ find the (zero-state response) if the input $x(t)$ is

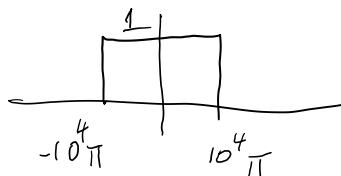
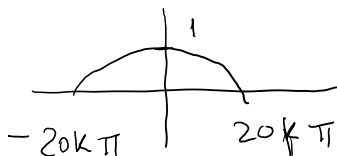
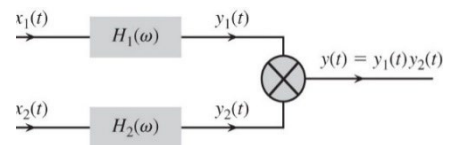
- a) $e^{-2t}u(t)$ b) $e^{-t}u(t)$ c) $e^t u(-t)$ d) $u(t)$

$$x_a(t) = [e^{-t} - e^{-2t}]u(t) \quad x_b(t) = t e^{-at} u(t)$$

$$x_c(t) = \frac{1}{2} e^{-t} u(t) + \frac{1}{2} e^t u(-t) \quad x_d(t) = (1 - e^{-t})u(t)$$

Q5) Signals $x_1(t) = 10^4 \text{rect}(10^4 t)$ and $x_2(t) = \delta(t)$ are applied at the inputs of the ideal low-pass filters shown below $H_1(\omega) = \text{rect}(\omega/40,000\pi)$ and $H_2(\omega) = \text{rect}(\omega/20,000\pi)$. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$.

- a) Sketch $X_1(\omega)$ and $X_2(\omega)$. b) Sketch $H_1(\omega)$ and $H_2(\omega)$.
c) Sketch $Y_1(\omega)$ and $Y_2(\omega)$. d) Find the bandwidths of $y_1(t)$, $y_2(t)$, and $y(t)$.



$$BW_{y_1} = 10 \text{ k}$$

$$BW_{y_2} = 5 \text{ k}$$

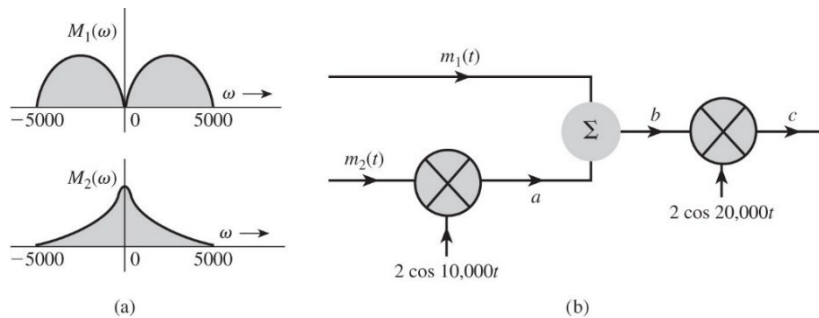
$$BW_y = 15 \text{ k}$$

Q6) For the signal $x(t) = \frac{2a}{t^2 + a^2}$ determine the essential bandwidth B (in hertz) of $x(t)$ such that the energy contained in the spectral components of $x(t)$ of frequencies below B Hz is 99% of the signal energy E_x .

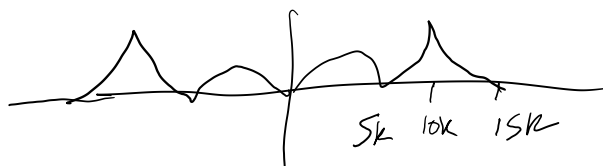
$$B = \frac{0.366}{a} \text{ Hz}$$

Q7) Figure below shows a scheme to transmit two signals $m_1(t)$ and $m_2(t)$ simultaneously on the same channel (without causing spectral interference). Such a scheme, which transmits more than one signal, is known as signal *multiplexing*. In this case, we transmit multiple signals by sharing an available spectral band on the channel, and, hence, this is an example of the *frequency-division* multiplexing. The signal at point b is the multiplexed signal, which now modulates a carrier of frequency 20,000 rad/s. The modulated signal at point c is now transmitted over the channel.

- a) Sketch the spectra at points a , b , and c . b) What must be the minimum bandwidth of the channel?
c) Design a receiver to recover signals $m_1(t)$ and $m_2(t)$ from the modulated signal at point c .



at node b



at node c

